

The Paradox of Plows and Productivity: An Agronomic Comparison of Cereal Grain Production under Iroquois Hoe Culture and European Plow Culture in the Seventeenth and Eighteenth Centuries

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Iroquois maize farmers in the seventeenth and eighteenth centuries produced three to five times more grain per acre than wheat farmers in Europe. The higher productivity of Iroquois agriculture can be attributed to two factors. First, the absence of plows in the western hemisphere allowed Iroquois farmers to maintain high levels of soil organic matter, critical for grain yields. Second, maize has a higher yield potential than wheat because of its C4 photosynthetic pathway and lower protein content. However, tillage alone accounted for a significant portion of the yield advantage of the Iroquois farmers. When the Iroquois were removed from their territories at the end of the eighteenth century, US farmers occupied and plowed these lands. Within fifty years, maize yields in five counties of western New York dropped to less than thirty bushels per acre. They rebounded when US farmers adopted practices that countered the harmful effects of plowing.

HISTORIANS, GEOGRAPHERS, AND ANTHROPOLOGISTS HAVE characterized the introduction of the plow as a critical step in agricultural development, enabling farmers to bring greater amounts of land under cultivation,

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to produce higher yields on those lands, and consequently to support higher population densities. But, before the arrival of Europeans, indigenous farmers in the Americas supported large populations with maize-based agriculture, implemented without plows. In Central Mexico alone, highly productive maize-cropping systems produced food for about eighteen million people at the time of European arrival. If plows were a critical step in agricultural development, how did maize farmers in the Americas, operating without animal-drawn tillage tools, achieve such high levels of agricultural productivity? Exploring this anomaly requires a more complex analysis of the role of plows and tillage in agriculture, one that includes an understanding of the biophysical factors that determine many agronomic practices such as crop selection, field preparation, planting densities and patterns, nutrient supply, pest management, and the scheduling of agricultural operations. These practices are all largely dependent on the biological and physical elements that comprise the environments in which farmers work. Using plows to prepare fields for planting fundamentally transforms an agricultural system; it puts in play an array of biological, physical, and chemical interactions that affect crop growth in complex ways, both favorably and negatively. When scholars assume that plowing is essential for productive agriculture, without an understanding of the underlying agronomic practices and processes that it affects, they may incorrectly conclude that agricultural systems without plows are inherently less productive. In fact, the productivity of indigenous farmers in the Western Hemisphere was a direct consequence of the *absence* of plows and tillage. An agronomic analysis that explains the role of plows challenges the assumption that farmers who lack them are at a disadvantage, with inherently less productive agricultural systems.¹

Comparisons of indigenous agriculture and European agriculture have been hampered by inaccurate suppositions. Many scholars assume that, with the exception of small pockets of intensive, permanent cropping (the American Southwest, Central America, and the Andes), the majority of indigenous farmers in the Western Hemisphere practiced shifting cultivation on small plots of land with infertile soils, located within large tracts of forest. They further reason that, without domesticated animals to provide manure to restore soil fertility, indigenous farmers had to rely on the regeneration of natural vegetation that was cut and burned to supply nutrients for crops. This characterization of indigenous

agriculture and its associated rationale, applied to the Americas as a whole, is largely incorrect. For Iroquois agriculture in eastern North America, it is altogether erroneous.²

The Western Hemisphere contains a vast array of environmental conditions; consequently indigenous farmers devised many different cropping systems to accommodate those varied conditions. They farmed in regions where water, its scarcity or overabundance, dictated most agronomic decisions, and in other areas where the length of the growing season was the primary factor. They farmed on the enormously fertile Mississippi Bottomlands and in nutrient-deficient soils in the Amazon Basin. They grew crops in regions where temperature and precipitation were relatively constant throughout the year, allowing two or three cropping periods, and in areas where dry/wet or cold/warm seasons limited harvests to one per year. In each environment they devised agricultural systems to meet the challenges and opportunities of that locale. As a result, indigenous agriculture in the Western Hemisphere was as varied as the environments in which it was practiced.

Consequently, a comparison of agriculture in Iroquoia and Europe must first consider the biophysical factors of each region. The two areas are remarkably similar in climatic and soil resources; both groups of farmers had access to large tracts of fertile soils in which temperature limited the growing season to one primary harvest per year. Surprisingly, farmers in Iroquoia, like their European counterparts, practiced large-scale, permanent agriculture, with extensive acreages under cultivation. Yet their agricultural systems were also markedly different. European agriculture depended on the integration of animal husbandry and cereal grain production, with plowing an essential component of the system. In contrast, the Iroquois used neither domesticated animals nor plows, although they too relied on a cereal grain as the foundation of their agriculture. While the Europeans grew wheat, the Iroquois planted maize.

Iroquoian farmers, without plows or draft animals, produced three to five times as much grain per unit of land area as their European counterparts. An agronomic analysis provides a framework for understanding this apparent anomaly. The explanation begins with the plow. Agricultural systems are extraordinarily complex; tillage, or its absence, affects soil properties, crop selection, weed control, and a host of other cascading interactions. In this comparison of Iroquois and European agriculture,

the characteristics of the crops themselves also play central roles as they interact with different methods of field preparation. Maize, inherently more productive than wheat, is uniquely suited to no-plow conditions. The agronomic practices of field preparation (plow versus no-plow) and the botanical characteristics of the crops (maize versus wheat) together explain these differences in productivity. Only by examining these agronomic and botanical factors can the two cropping systems be constructively compared and the differences in yields explained.

Iroquoian and European agricultural systems were situated within very different social, political, cultural, and economic environments, but they operated within a remarkably similar physical environment. Although scholars have focused primarily on the dissimilarities in agriculture in these two regions, North American agriculture in the seventeenth and eighteenth centuries shared more features with European agriculture than it did with other systems of indigenous agriculture in tropical America. Even today, North Americans farm within an environmental context that resembles Europe far more than other parts of the Americas.

The Iroquoian homelands and Western/Central Europe are both located in the humid temperate zone; Europe has a maritime temperate climate while Iroquoia falls within the continental temperate zone. Both regions lie between the 40th and 55th north latitudes, with periodic fluctuations in day length throughout the calendar year, and both have a single annual growing season defined by temperature and day length, rather than rainfall. Rainfall distribution and temperature regimes in both areas are suitable for a wide range of temperate crops. The most productive agricultural soils of the northern United States are alfisols, inceptisols, and mollisols, all relatively young soils that have been recently glaciated and are characterized by high-base status, making them moderately or very fertile. These same recently glaciated and fertile soils are found in the major agricultural regions of Europe. The number and timing of growing seasons in a calendar year, along with the distribution of rainfall determines a region's agricultural profile, shaping crop selection and agricultural practices; soil fertility determines the area's agricultural potential. Consequently, Europe and Iroquoia shared a remarkably similar resource base for agriculture.³

As noted above, anthropologists, geographers, economists, and historians have characterized tillage as fundamental to agriculture and have

equated the adoption of plows, almost universally, with increased agricultural productivity. Within the United States, agricultural scientists from the nineteenth through the first half of the twentieth century described plowing as a critical practice for successful crop production. More recently, however, agricultural scientists and some historians have regarded tillage with considerable ambivalence, acknowledging its beneficial aspects, but expressing increased concern with the associated soil erosion and compaction, and the negative effects on soil organic matter and tilth. Most US agricultural scientists today recommend that farmers significantly decrease the number and intensity of tillage operations. In fact, many farmers have transitioned to cropping systems that use no tillage implements at all or use specially designed plows that minimize soil disturbance and leave most plant residues on the surface. Collectively these systems are called conservation tillage practices. The history of tillage and its agronomic effect on European agriculture help explain the discrepancy between these perspectives in which some view plowing as the mark of agricultural advancement, and others urge farmers to abandon plows.⁴

Viewed broadly, historians have attributed increases in European grain production beginning in the early Middle Ages to a four-step process. During the early Middle Ages European farmers used a two-field system in which crops were first seeded to cereal grains (wheat, oats, rye, or barley); the following year the field was rotated to pasture. They developed improved plows that included a coulter for cutting sod, a plowshare that cut the soil horizontally at rooting depth, and a moldboard that inverted the cut sod. These improved plows allowed farmers to cultivate the heavier soils found in central, western, and northern Europe that had been unworkable with the lighter ard. The introduction of the horse collar and harness enabled horses to draw the heavier plows more quickly, leading to larger areas under cultivation and increased cereal production. Animals were critical to the system, providing power for tillage while also producing manure that could be used to increase fertility on fields planted to cereal grains.⁵

Later in the Middle Ages the three-field system replaced the biennial rotation. A winter grain was followed by a short season crop of turnip or bean, which was in turn succeeded by a spring grain, reducing the fallow land to just one third of a farm's acreage. The three-field system increased

the number of animals that could be maintained on a farmstead and the amount of manure applied to fallow lands. With the introduction of short season fodder crops and grain legumes, farmers were able to fit an additional crop into the rotation and thus expand the number of livestock and improve human diets. The expanded use of domesticated animals allowed farmers to increase the area of land under cultivation, increase yields on those fields through enhanced fertility gained from better use of their manure, and improve the lives of farm families with additions of dairy, meat, and legumes in their diets. It was this combination of tighter integration of livestock with cropping, and the more effective use of tillage tools that allowed grain yields in Europe to double during the Middle Ages; but even with these gains, wheat yields seldom surpassed fifteen bushels per acre. Yields of twelve to fifteen bushels per acre were typical of much of Europe through the eighteenth century and later.⁶

In the nineteenth century Europeans began to systematically incorporate clover and alfalfa into their cropping systems. The Norfolk rotation, which included wheat, clover, oats, and turnip, enhanced soil fertility within the cropping system, with wheat yields in Europe rising above fifteen bushels per acre. Finally, intensive, modern agricultural practices (inorganic fertilizers, improved varieties, and pesticides) adopted in the latter part of the twentieth century allowed European farmers to produce grain yields that now range from sixty to one hundred twenty bushels per acre.⁷

Agronomists view this history of increased wheat yields in Europe much differently. Although the use of improved plows enabled farmers to expand agriculture into areas where soils were too heavy to cultivate with lighter plows, intensive and continuous tillage of agricultural lands ultimately created more problems than it solved. Over the long term, plowing reduces soil fertility and physically degrades soils to the point where grain yields inevitably decline. It was only with the introduction of forage legumes in the nineteenth century that European farmers were able to ameliorate these problems and develop sustainable wheat production systems.

Tillage provides many advantages. Plowing soils prior to planting achieves several objectives critical for growing successful grain crops; it provides a seed bed with appropriate moisture that facilitates good soil/seed contact, an unrestricted rooting zone, and an area free of competing

weeds. Small-seeded grain crops like wheat, barley, rye, and oats need a finely prepared seedbed to facilitate germination. Each seed must be in direct, firm contact with moist soil in order to take up the water that initiates seed germination. If the seed receives enough moisture to start germination, but then exhausts soil moisture in the immediate area before its young main root is strong enough to push through the soil to exploit a larger volume of soil (and water), it will die before emerging or shortly thereafter. Early farmers broadcast their small-seeded cereal grains by hand on soil that had been lightly disturbed to provide these conditions. If they planted their small seeds on unworked soil, the seeds would either not germinate because they could not take up sufficient moisture or they would germinate but then die because the young roots could not penetrate the soil before its seed reserves and available moisture were exhausted. Farmers probably discovered quickly that there was fine line between preparing soils that were too loose and those that were too compacted. Seeds initially need firm contact with soil to take up water, but if the soil is too compacted, the young roots cannot penetrate the hard soil. Tillage operations play a key role in obtaining these optimal conditions.⁸

Working or tilling the soil before planting also removes existing vegetation that can out-compete the struggling seedlings. Plants with established roots systems have first access to available water and nutrients in soil, and their leafy canopies intercept incoming light energy, all at the expense of small seedlings at ground level. Removing weeds before sowing thus levels the growing field for small-seeded grain crops.

When farmers plow with animals, they affect the cropping system in complex and subtle ways. Farmers in the late Middle Ages learned to stable or fold animals, enabling them to concentrate the nutrients from manure directly on land used to grow cereals. Plowing these fields incorporated the manure, facilitating its decomposition and the release of inorganic nitrogen immediately available to plants. Thus, farmers achieved significant increases in cereal yields from a combination of small practices associated with the care and use of draft animals. But animals are not uniformly advantageous for agriculture. Draft animals must be fed; as much as one third of the land in pre-modern Europe was planted to fodder and forage crops for livestock.⁹

In terms of soil fertility, plowing can be advantageous because it increases the oxidation of soil organic matter by exposing more of the

soil to air. When this organic matter oxidizes, it releases nutrients in forms that are available to plants. Consequently, when a pasture or meadow is first plowed, it provides a flush of inorganic nitrogen that can be immediately used by cereal grains planted in the former pasture. As nitrogen is usually the most limiting nutrient for crop growth, grain yields will increase dramatically in these fields. When farmers lack other sources of nutrients for plant growth, plowing fields that have been fallowed and allowed to accumulate soil organic matter is a reliable mechanism for providing nitrogen to their grain crops.¹⁰

There are drawbacks to tillage as well. This mechanism of oxidizing soil organic matter to release nitrates does not work indefinitely. Large amounts of nitrates are released the first year, but in subsequent years, when the field is repeatedly plowed, soil organic matter levels decline, and less nitrogen becomes available each year. Grain yields inevitably decrease because eventually there is insufficient nitrogen for crop growth. When fields with moderate to high levels (3 to 4 percent) of soil organic matter are first plowed, organic matter levels decline rapidly in the first years of cultivation until they reach a stable lower value of 2 percent (usually within two to four decades), which is very resistant to further oxidation. Soils with these lower levels of organic matter present many problems for crop growth: they have poor structure and decreased water-holding capacity; they are more prone to compaction; and they have less ability to provide nutrients to crops. Plowing also makes soil enormously vulnerable to erosion in the weeks after plowing and before the new crop has established an effective leaf canopy to protect the soil from wind and water. Even fields with small slopes can lose large amounts of topsoil that contain most of the organic matter, further damaging the soil's ability to produce crops.¹¹

European farmers eventually solved the problem of declining soil organic matter associated with plowing in the nineteenth century; they incorporated *forage* legumes as part of the Norfolk rotation. Although all leguminous crops can add nitrogen to the cropping system through the symbiotic association of bacteria on their roots, most of the nitrogen from *grain* legumes such as bean, pea, and lentil, is removed when the seeds are harvested and eaten by people. These crops are grown in widely spaced rows; the amount of plant residue left after harvest is relatively small. Most grain legumes are also annuals; once they set seed, the plants

do not accumulate more biomass and their root systems die. Although the introduction of grain legumes (as part of the transition from a two-field to three-field system) in the late Middle Ages provided additional nitrogen, neither the grain legume nor the additional manure obtained from larger numbers of livestock was sufficient to boost grain yields above twelve to fifteen bushels per acre. Forage legumes were needed to provide more nitrogen to the system.¹²

Forage legumes, in contrast to grain legumes, are seeded close together and once established, they cover most of the ground. Because they are perennials, they continue producing biomass as long as there is sufficient warmth and water. Forage legumes can be harvested several times a year because their roots systems do not die after setting seed; they immediately begin producing additional top and root growth. In fact, very large amounts of dry matter from the stubble and roots remain even after the forage has been cut and removed for animal feed. Consequently, perennial forage legumes typically produce four to eight times as much dry matter in their stubble and root systems as do grain legumes and two to three times as much nitrogen for succeeding crops. In addition, when forage legumes are then fed to livestock, a significant portion of plant nitrogen is later returned to cropland in their manure. (When people consume grain legumes, the nitrogen contained in the grain is lost permanently unless human waste is applied to cropland.)¹³

Forage legumes have other benefits that enable them to counter some of the negative effects of plowing. Because they are perennials, forage legumes take longer to establish; once farmers have a good stand, they are reluctant to plow them down and plant another crop. Farmers keep stands of alfalfa or clover for two or more years, eliminating tillage in those years. They continue to harvest hay for their livestock, so the land contributes to farm productivity. Thus, fields sown to a forage legume provide an extended period for soils to accumulate organic matter after cereal grain production. The large root systems of these forage legumes improve soil structure, and when they are finally plowed down, they not only release a large amount of inorganic nitrogen for the next crop, but they leave the soil in excellent physical condition for plant growth. Consequently, in order to establish continuous, productive, and stable cropping systems, European farmers had to successfully integrate forage legumes into their cropping systems to mitigate the negative effects of

plowing and the declining organic matter levels that come with repeated tillage of agricultural land. This did not occur widely across Europe until the nineteenth century and even later.¹⁴

Before colonization, indigenous farmers in the Western Hemisphere did not have domesticated animals that could be used to pull plows. Instead, they devised very different cropping strategies that enabled them to successfully grow crops without plowing. Until quite recently, scholars identified two major cropping systems present in the Americas on the eve of colonization, neither of which utilized plows: intensive, permanently cropped fields and shifting cultivation. Permanent cropping systems, which appeared quite limited in their extent, used soil amendments, terracing, irrigation, and intermittent flooding to maintain soil fertility and manage weeds; extended fallow periods were absent. The highly productive maize systems of Central America, Peru, and the American Southwest are examples of permanently cropped agriculture. In contrast, scholars thought that shifting cultivators farmed the majority of forested land areas in the Americas. They cut and burned primary or secondary forest, planted their crops in unplowed soil with digging sticks or hand hoes, and then allowed the fields to rest or lie fallow for periods that exceeded the cycles of crop production. During these long fallow periods, plant biomass accumulated; when the vegetation was subsequently cut and burned, it provided nutrients for the next cycle of crop growth.¹⁵

But several researchers have challenged this dichotomy of permanent, intensive agriculture versus fallow-based farming. In particular they have questioned the assumption that indigenous farmers in forested lands practiced shifting cultivation to the exclusion of more intensive forms of agriculture. In Amazonia, the region most typically associated with shifting cultivation because of its large areas of nutrient-deficient soils, William Denevan asserts that archeological, ethnographic, and ecological evidence indicates that pre-Columbian farmers engaged in a mosaic of cropping practices, many of which involved intensive, permanent management of agricultural fields. He argues that shifting cultivation, rather than representing an ancient agricultural practice in this region, probably emerged as an historic adaptation to changing conditions under colonization. In Mesoamerica, researchers for many decades assumed that ancient Mayan farmers engaged almost exclusively in shifting cultivation and

that practices of contemporary Mayans were similar, if not identical to, those of pre-Columbian Mayans in the region. But more recent research has now firmly established that earlier Mayan farmers practiced intensive, permanent agriculture.¹⁶

In eastern and central North America many scholars have argued that, with the exception of continuous cropping in fertile bottomlands, most indigenous farmers used fallow-based cropping systems. But more recent research emphasizes the presence of continuously cropped fields in many areas of North America. William Doolittle, a geographer who has catalogued and summarized the ethnographic, historic, and archeological evidence for shifting cultivation in North America, states, "the overwhelming corpus of evidence indicates clearly and convincingly that although native North Americans cleared the forest by slashing and burning, they did so for permanent clearance."¹⁷

For decades scholars typically described the Iroquois as shifting cultivators, basing that characterization primarily on reports by Joseph-Francois Lafitau, Gabriel Sagard, and Samuel de Champlain, who described Huron farming practices in the seventeenth and eighteenth centuries. According to these reports, Huron farmers in what is present-day Ontario and Quebec periodically abandoned their fields because of declining soil fertility. Although the Huron spoke a Northern Iroquoian language and were geographically close to the Iroquois Confederacy (which included the Mohawk, Seneca, Onondaga, Oneida, Cayuga, and later, Tuscarora nations), they farmed in areas with a shorter growing season and less fertile soils.¹⁸

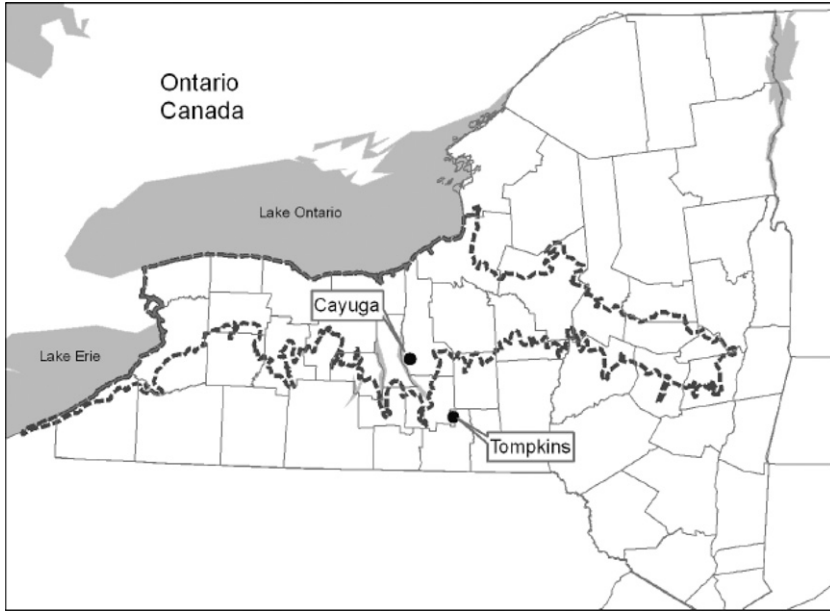
In contrast, maize-based agriculture provided a major portion of the Iroquois diet. They planted large fields of maize south of Lake Ontario on alfisols and inceptisols, some of the most productive soils in the Northeast. Nutrient-deficient soils are a primary rationale for farmers to use shifting cultivation, but Iroquois soils were hardly infertile. In addition, neither ethnographic nor historical records indicate that Iroquois farmers in these areas cut and burned or fallowed their fields. While there is little evidence to characterize the Iroquois as shifting cultivators, there is documentation to show that they were engaged in a highly productive agriculture that is most accurately described as a continuous cropping.¹⁹

From at least the fifteenth century, maize was the centerpiece of agriculture among the nations that comprised the Confederacy. Three

Sisters—intercropped maize, beans, and squash—were a dominant factor in Iroquoian social, political, cultural, and economic spheres when Europeans arrived in the sixteenth century in what is present-day New York and Ontario. Historical and ethnographic accounts from English, French, and Dutch colonists provide extensive evidence that the Iroquois and their neighbors implemented agriculture across large acreages of land and produced significant quantities of maize grain from their agricultural fields.²⁰

Two descriptions of Seneca agriculture, separated by almost one hundred years, are helpful in understanding Iroquoian farming techniques: the first a French account from 1687 and the second from the 1779 Sullivan Campaign during the Revolutionary War. Both describe attacks by colonial forces against Seneca towns located in the Ontario Lake Plain, the most agriculturally productive region of present-day New York, that extends across the center of the state (see Figure 1). During the last glaciation, limestone deposits found in a narrow band from Buffalo to Albany were first exposed and then scraped southward, creating a swath of fertile soils from Albany to Buffalo. Seneca, as well as other Iroquois farmers, concentrated their agricultural activities in this band of highly productive soils beginning in the fifteenth century.²¹

In the summer of 1687 the Marquis de Denonville, French governor of New France, marched from Montreal into central Iroquoia with a force numbering more than two thousand, destroying Seneca villages and their adjoining cornfields just south of present-day Rochester. His attack was undertaken because the Seneca had allied themselves with the English, who contested France for control of North America. Denonville encountered large quantities of stored grain in the Seneca towns, as well as an abundance of maize still growing in the fields surrounding these communities. In describing the campaign, he wrote that his army “destroyed a vast quantity of fine large corn, beans, and other vegetables of which there remained not a single field, and after having burned so large a quantity of old corn that the amount dared not be mentioned.” He also wrote, “The quantity of corn which we found in store in this place, and destroyed by fire is incredible.” By the conclusion of the campaign, he boasted of having destroyed more than 1.2 million bushels of maize, a combination of maize stored in granaries and on plants still in the field. Denonville was clearly astonished at both the extent and productivity of

Figure 1. Ontario-Erie Plain and Finger Lakes Region.

Area within dotted line shows location of most agriculturally productive soils in present-day New York State. Cayuga and Tompkins refer to counties where field experiments identified in the text were conducted.

SOURCE: Map by Stephen D. Smith.

Seneca agriculture, perhaps comparing it to his own knowledge of small-grain production in Europe at the same time.²²

Almost one hundred years later, another army was ordered by George Washington to destroy and disrupt Seneca and Cayuga towns in retaliation for their support of Great Britain during the Revolutionary War. Washington directed John Sullivan, the general in charge of the campaign, to specifically target agricultural fields: "It will be essential to ruin their crops now in the ground and prevent their planting more." Over the course of a six-week campaign in the summer of 1779, Sullivan's army destroyed more than forty Seneca and Cayuga villages and their agricultural fields in a large swath across the Finger Lakes region of New York. He claimed to have destroyed 160,000 bushels of maize during the campaign. Many officers in Sullivan's army recorded in their diaries and journals detailed descriptions of the agricultural fields they burned.²³

John Burrowes's entry on August 27, 1779—one of dozens—suggests the surprise and awe of American colonists when they encountered Iroquois maize fields. "We got this night at a large flat three miles distant from Chemung where corn grows such as cannot be equaled in Jersey. The field contains about 100 acres. beans. cucumbers. Simblens water-melons and pumpkins in such quantities (were it represented in the manner it should be) would be almost incredible to a civilized people." Lieutenant Samuel Shute, on September 7, wrote, "It [town of Seneca Castle] is situate 2 miles from the Seneca Lake which is thirty five miles long and in some places 10 broad & the most rich and fertile country all round it—We found 200 acres of exceedingly good corn intermixed with beans, & squashes pompions & a few potatoes." And again on September 15, he reported, "The army was employed in destroying corn, which was not less than 200 acres intermixed with beans and the best I ever saw."²⁴

Neither Denonville's reports or the journal entries from the Sullivan campaign described fallowed or abandoned lands or the cutting and burning of agricultural fields, telling characteristics of shifting cultivation. Rather, these reports portray a long-term, continuous cropping system that was not only productive but also persistent. Many of the fields that soldiers in Sullivan's army described were located less than fifty miles west of the villages where Denonville had cut down vast acreages of corn almost one hundred years earlier.

The productivity of different agricultural systems can also be assessed in other terms. Wheat and maize grain provide similar amounts of energy and fat. Although wheat contains more protein than maize, like quantities of each are needed to satisfy the caloric needs of people and animals. Consequently, comparing wheat and maize yields indicates the number of people or animals that can be supported per unit of land and provides a measure by which the productivity of each can be evaluated. Although records of European wheat yields in the seventeenth and eighteenth centuries are incomplete, historians have compiled sufficient information from an array of sources to provide a reasonably accurate estimate. Manorial accounts, tithe records, probate inventories, journals, diaries, and other informal records of both farmers and travelers have been mined for their indirect information to provide a broad and realistic assessment of wheat yields. Table 1 provides estimates of wheat yields for England and Western Europe in the seventeenth and eighteenth centuries.

Table 1. Productivity of Wheat.

Time Period	England		England		Germany		The Netherlands		France	
	Van Bath		Turner et al.		Van Bath		Van Bath		Van Bath	
	YSR	bu acre ⁻¹	YSR	bu acre ⁻¹	YSR	bu acre ⁻¹	YSR	bu acre ⁻¹	YSR	bu acre ⁻¹
1601–1650	7.20	18.00			3.60	9.00				
1651–1700					5.60	14.00				
1701–1750			8.40	21.00	6.00	15.00				
1751–1800	10.40	26.00	8.00	24.00			21.20	53.00	6.40	16.00

Productivity of wheat expressed as yield seed ratios (YSR) and bushels per acre from various countries in Europe in the seventeenth and eighteenth centuries.
SOURCES: Slicher van Bath, *Agrarian History*, 330–33; Turner et al., *Farm Production in England*, 163, 167.

These estimates are summarized in fifty-year increments, averaging the various values but without weighting the values for the number of observations because the data is incomplete. Data reported as yield to seed ratios (the number of bushels harvested per bushel seeded) assumes a seeding rate of 2.5 bushels per acre to convert the ratios to yields per acre. Table 2 provides combined yield estimates of wheat, rye, and barley for three groups of countries from 1600 to 1799 with more than nine thousand individual ratios used to construct the table. The original data was in yield seed ratios from which were calculated yields per acre using a seeding rate of 2.5 bushels per acre.²⁵

England and the Low Countries (the Netherlands, French Flanders, and Belgium) had the highest levels of grain productivity, with yields ranging from seventeen to twenty-five bushels per acre and yield seed ratios from 6.8 to 10.4. Grain productivity in southwestern Europe (France, Spain, and Italy) was somewhat lower with yields of sixteen to eighteen bushels per acre and seed yield ratios from 6.4 to 7.2. Yields in northern Europe (Germany, Switzerland, and Scandinavia) ranged from nine to fifteen bushels per acre with yield seed ratios of 3.9 to 6.0. Grain productivity in eastern Europe was lowest; yields were between 8.8 and 11.8 bushels per acre and yield seed ratios between 3.5 and 4.7. These data represent means and thus do not reflect the higher yields that individual farmers obtained on better soils with more intensive management. For example, Dutch historian, B. H. Slicher van Bath reported a yield

Table 2. Productivity of Combined Wheat, Barley, and Rye.

	<i>Temperate North Sea</i>		<i>South and Western Europe</i>		<i>Northern Europe</i>		<i>Eastern Europe</i>	
	YSR	bu acre ⁻¹	YSR	bu acre ⁻¹	YSR	bu acre ⁻¹	YSR	bu acre ⁻¹
Time Period								
1600–1649	6.80	17.00			4.40	11.00	4.00	10.00
1650–1699	9.20	23.00	6.40	16.00	4.00	10.00	3.80	9.50
1700–1749			6.40	16.00	4.00	10.00	3.50	8.80
1750–1799	10.00	25.00	7.20	18.00	5.20	13.00	4.70	11.80

Productivity expressed as yield seed ratios (YSR) and bushels per acre.

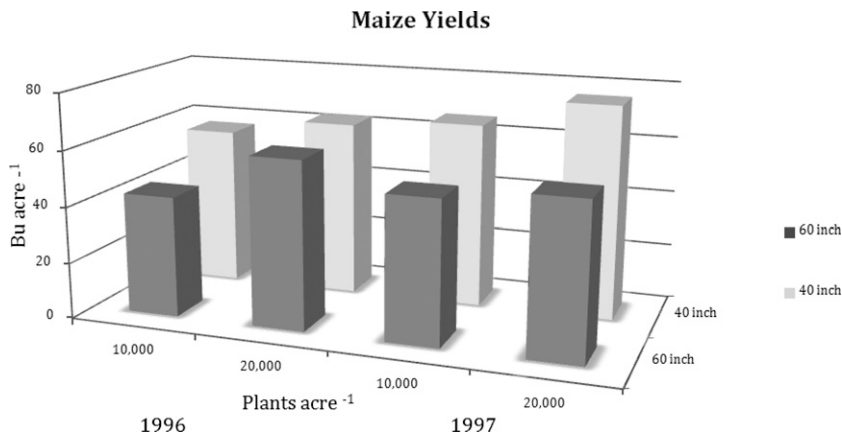
SOURCE: B. H. Slicher van Bath, "Agriculture in the Vital Revolution," in *The Cambridge Economic History of Europe: Vol. 5, The Economic Organization of Early Modern Europe*, ed. E. E. Rich and C. H. Wilson (Cambridge: Cambridge University Press, 1977), 81.

seed ratio of 20.2 (fifty-one bushels per acre) in Harwell, England, between 1612 and 1620 and yield seed ratios of fifteen to twenty-eight (thirty-eight to seventy bushels per acre) in the Netherlands between 1765 and 1779. England and the Low Countries were noted for their superior agriculture and higher cereal grain yields compared to the rest of Europe. Van Bath attributed higher yields in the Low Countries to heavy fertilization (animal manure, night soil, peat ash, and lime). But broadly speaking, European wheat productivity in the seventeenth and eighteenth centuries was meager and seldom bountiful; farmers typically harvested less than eight seeds for each seed they planted.²⁶

The accounts by Europeans, Euroamerican colonists, and historians described above provide anecdotal and qualitative information about Iroquois maize productivity, but no quantitative estimates are found in the historical records. However, recent field research suggests that Iroquois farmers in the seventeenth and eighteenth centuries produced maize grain yields of twenty-two to seventy-six bushels per acre. Yields of forty-three to seventy-six bushels per acre were produced on the highly productive soils of the Ontario Lake Plain. These experiments were established with a maize variety and agricultural conditions similar to those used by the Iroquois at the time. The experiments, conducted over a four-year period at two different sites, were designed to determine yield levels of maize, bean, and squash in two different soil/climatic regions of New York located in Cayuga and Tompkins counties. The replicated treatments included variables for cropping systems, maize population, and maize spacing. Over two years and two locations, maize was harvested from more than sixty-five individual three-meter square plots. These experiments provide the most extensive and reliable empirical estimates for Iroquoian maize yields. Furthermore, they confirm the observations of historical and ethnographic documents that describe an enormously productive agriculture in the region.²⁷

Figure 2 provides a summary of the results of the field experiments on the Ontario Lake Plain. The highest yields were obtained when maize was planted at a narrower spacing (forty versus sixty inches) and at a higher population (twenty thousand versus ten thousand plants per acre). Table 3 provides the yield data as bushels per acre and as yield seed ratios. Approximately ten and twenty pounds of corn seed per acre would be required to obtain plant populations of ten thousand and twenty

Figure 2. Yields of Iroquois Maize.



Maize yields using a traditional variety and practices, as affected by plant population and spacing, Cayuga County, New York, 1996–1997.

SOURCE: Mt. Pleasant and Burt, “Estimating Productivity,” 52–79.

thousand plants per acre, respectively. This assumes that a farmer would plant about 10 percent more seed than needed for the desired final population in order to account for losses from poor germination, predation, etc. Seed yield ratios in these experiments ranged from 154 to 368.

Wheat yields from Europe, as a whole, and from field experiments with Iroquois maize have been separated into three levels (low, medium,

Table 3. Productivity of Maize.

Year	Spacing (inches)	Population (plants acre ⁻¹)	bu acre ⁻¹	YSR
1996	40	10,000	57	320
		20,000	63	176
	60	10,000	43	242
		20,000	60	168
1997	40	10,000	66	371
		20,000	76	213
	60	10,000	51	287
		20,000	55	154

Productivity of maize as affected by spacing and plant population from field experiments in Cayuga County, New York, 1996–1997, expressed as bushels per acre and yield seed ratios (YSR).

SOURCE: Mt. Pleasant and Burt, “Estimating Productivity,” 52–79.

and high) to show the range of yields and the associated seed yield ratios at each level. Iroquois farmers harvested three to five times as much grain per unit land area as did European farmers, with seed yield ratios that were orders of magnitude greater (see Table 4). The greater productivity of Iroquois farmers resulted from the absence of plows, which forced them to adopt cropping practices that conserved soil organic matter, their most valuable soil resource. But the Iroquois also benefited from their use of maize, with its higher yield potential compared to wheat, and from its other botanical characteristics, which enabled maize to provide a much greater return on investment, particularly in no-plow conditions.

Lack of nitrogen was the primary constraint to higher European wheat yields in the seventeenth and eighteenth centuries. Soil with 2 percent organic matter would provide less than forty-five pounds per acre of nitrogen from the oxidation of soil organic matter annually. Other sources of available nitrogen from decomposition of small amounts of leguminous residues and animal manure, and nitrogen contained in rainfall and atmospheric dust, might contribute an additional fifteen pounds per acre, providing a pool of sixty pounds per acre of nitrogen for crop growth. But crop plants are unable to fully use all available nitrogen, with significant plant-available nitrogen lost to leaching, volatilization, erosion, nitrogen immobilization, and uptake by weedy plants. Consequently, depending on the conditions of individual fields, twenty to twenty-five pounds per acre, or even less, of the nitrogen would be available for uptake by wheat. Since wheat grain contains 2 percent nitrogen, fifteen bushels per acre of

Table 4. Comparisons of Three Levels of Maize and Wheat Productivity.

<i>Yield level</i>	<i>Wheat</i>		<i>Maize</i>	
	<i>bu acre⁻¹</i>	<i>YSR</i>	<i>bu acre⁻¹</i>	<i>YSR</i>
Low	8.00–12.00	3.50–4.80	40.00–49.00	112.00–137.00
Moderate	13.00–17.00	5.20–6.80	50.00–59.00	140.00–165.00
High	18.00–25.00	7.20–10.00	60.00–76.00	168.00–196.00

Expressed as bushels per acre and yield seed ratios (YSR).

SOURCES: Mt. Pleasant and Burt, "Estimating Productivity," 52–79; Slicher van Bath, *Agrarian History*, 330–33.

wheat removes eighteen pounds of nitrogen per acre. In order to increase wheat yields above fifteen bushels per acre, European farmers had to find other ways to add nitrogen to their agricultural systems. They found the solution by integrating clovers and alfalfa into their cropping systems. These crops not only improved soil organic matter levels on worn-out lands, they also enabled farmers to increase their livestock numbers, which produced more manure that could be used to fertilize grain fields. But most European farmers did not implement legume-based rotations until the nineteenth and twentieth centuries.²⁸

Iroquois maize farmers operated within a very different agricultural setting. Instead of plowing, they used hand tools to work only those portions of their fields where they planted individual (or small groups) of maize kernels. Since Iroquois fields were not plowed, soil organic matter levels did not decline as they did in European cropping systems. Instead, Iroquoian farmers could rely on the slow, but steady oxidation of soil organic matter to release plant-available nitrogen. Soils in the Ontario Lake Plain are inherently fertile in their uncultivated state, with significant amounts of soil organic matter; these unplowed alfisols could be expected to have 3.5 to more than 4 percent organic matter.²⁹

Soils with 4 percent organic matter can release ninety pounds per acre of plant-available nitrogen annually. These soils would have a much larger portion of their organic matter in the “active fraction,” which more readily releases nitrogen compared to soils that have been continuously plowed and whose organic matter contents have been reduced to 2 percent or less. Small additions from the residues of grain legumes, along with nitrogen inputs from precipitation and atmospheric dust would contribute another thirteen to eighteen pounds per acre of nitrogen. Losses from erosion would be minimal since the fields were not plowed. Furthermore, many fields adjacent to the major rivers in the Ontario Lake Plain experience intermittent flooding, periodically replenishing the soil with plant nutrients. Grain yields of fifty to seventy-five bushels per acre of maize contain approximately fifty to seventy-five pounds per acre of nitrogen in the grain. This could be supplied by plant-available nitrogen in the soil almost indefinitely, as long as grain yields remained in this range and the soils were not intensively plowed. Thus, the Iroquois had an inherently stable system that enabled them to harvest modest, but significantly larger, grain yields than European farmers.³⁰

Iroquois farmers also benefited from growing a more productive cereal grain than European farmers. Maize typically yields almost twice as much grain as wheat when grown in similar environments and under similar management systems. These yield differences have been consistent over long periods of time and over wide geographic areas. The yield advantage of maize comes primarily from two factors: its photosynthetic pathway and lower protein content. Maize uses a C4 pathway that is more efficient than the C3 pathway of most cereal grains, including wheat. C4 plants have higher rates of photosynthesis because they are able to concentrate carbon dioxide at higher levels and because they have lower losses from photorespiration. Plants with higher protein in their grains will have lower grain yields because the plants use more energy to produce proteins than to produce carbohydrates.³¹

Maize seeds are significantly larger than wheat. One pound of maize contains about twelve hundred seeds while one pound of wheat has ten times that number. Because of its larger seeds, maize does not require a finely prepared seedbed to facilitate soil-to-seed contact and germination. In fact, maize is particularly well-suited to the conditions of roughly prepared seed beds found in no-till conditions, in contrast to most other cereal grains. The large maize endosperms have more nutrients and the seeds take up more water during germination than do small seeds. As a result, they are more likely to survive and emerge as seedlings if conditions for early growth become unfavorable. Because their food reserves allow them to continue to grow, even if they are not able to begin photosynthesis immediately, the seedlings are hardy and less vulnerable to an unfavorable environment early in their development. Maize grows very rapidly above and below ground, in both its seedling stage and thereafter, giving it a significant advantage over weeds. Other cereal grains are very vulnerable to weeds because their small size does not allow them to compete as effectively early in the growing season, and they are less vigorous competitors even when fully grown.³²

The larger seed size of maize also allows farmers to plant the kernels individually or in small groups, in regular patterns with fixed plant populations, that optimally utilize incoming light energy, water, and nutrients. But wheat, because of its small seed size, is broadcast seeded (before the introduction of grain drills), resulting in irregular spacing and differential plant populations across the field. Although wheat plants can tiller (send

up additional stems and seed heads) when they detect large openings with their nearest neighbors, the resulting plant population across the field will be irregular in both spacing and number, with resources (light, water, and nutrients) less efficiently allocated to individual plants.³³

The large seed size of maize confers another advantage: the distance between individual plants or groups of plants allows the maize to be weeded during the growing season, reducing yield losses. Broadcast wheat, however, is too closely and irregularly spaced to permit weeding once the field is seeded; consequently, it can be seriously damaged by weeds that emerge later in the season. Maize is also less vulnerable to insects and diseases compared to wheat, which suffers significant losses from a wide array of pests and pathogens. Maize's larger kernel size may also give it an advantage in farmer breeding efforts. When farmers broadcast their wheat seed, individual seeds can land anywhere on the field. Throughout the growing season they manage the wheat as a population, not as individual plants. By contrast, farmers plant individual seeds of maize (rather than broadcast seeding a handful) and can tend maize plants as individuals or small groups in carefully spaced patterns. In selecting seeds for the next year's planting, maize farmers can identify specific plants with desirable characteristics. Consequently, they may more easily select maize plants that are adapted to specific microenvironments.³⁴

All these factors described above result in a much higher seeding rate for wheat since fewer of the seeds are likely to survive and produce mature kernels. Wheat in the seventeenth and eighteenth centuries was typically seeded at sixty to one hundred eighty pounds per acre, compared to maize, which required just ten to twenty pounds per acre. These differences are further magnified because of the lower yield potential of wheat. Since wheat produces less grain per unit land area and requires more grain to seed a field, the ratio of harvested grain to the amount seeded (yield seed ratio) is a mere fraction of that of maize. In Europe during the seventeenth and eighteenth centuries, one pound of wheat seed typically produced five to twenty pounds of harvested grain, while one pound of maize in our field experiments yielded one hundred fifty to more than three hundred pounds of grain. Wheat growers were further penalized, particularly in years with poor yields, because the portion of seed that had to be saved for the coming season was much greater than with maize. If growers harvested ten bushels per acre or less of wheat, at

least 20 percent of the crop would have to be set aside. But if maize yielded as little as ten bushels per acre, only 2 to 4 percent of the harvest would need to be saved for the coming year (see Table 5).³⁵

European wheat growers also had to provide feed for their animals by devoting part of their land to pasture and/or feeding them a portion of the harvested grain, which further decreased their net yields. Iroquois maize growers, because their cropping system did not include animals, could devote all of their fields to food production and consume their entire harvest, except that reserved for next year's seed.

Weed pressure in Iroquois territories in western New York was probably considerably less than in European fields in the same time period for two reasons. First, the absence of tillage facilitated weed management. Although plowing removes established weedy vegetation, it also brings to the surface weed seeds that can then germinate and compete with seedling wheat. Flushes of weed seedlings that follow plowing can be devastating to emerging small-seeded cereal grains. In fact, European farmers relied on a season of widely spaced grain legumes or other row crops, which, unlike cereal grains, allowed them to cultivate between the rows, to reduce weed pressure in their fields before they planted their cereal grains. Iroquois farmers did not face these recurring flushes of

Table 5. Portion of Wheat and Maize Yield Saved for Seed.

	<i>Wheat</i>			<i>Maize</i>		
	<i>Seeding Rate (lbs acre⁻¹)</i>					
	60.00	120.00	180.00	8.00	12.00	16.00
<i>Grain Yield (bu acre⁻¹)</i>	<i>Percent Yield Saved for Seed</i>					
10	10.00	20.00	30.00	1.40	2.10	2.80
15	7.00	14.00	21.00	<1.00	<1.00	<1.00
20	5.00	10.00	15.00	<1.00	<1.00	<1.00
25	4.00	8.00	12.00	<1.00	<1.00	<1.00
30	3.00	6.00	9.00	<1.00	<1.00	<1.00
50	2.00	4.00	6.00	<1.00	<1.00	<1.00

Portion of wheat and maize yield saved for seed at different yield levels and under varying seeding rates.

weed seedlings because they were not plowing; they were also able to more easily remove weeds in their maize fields because of the space between the mounds of maize plants.³⁶

Second, Iroquois farmers faced weeds that were less pernicious and aggressive than weeds in European fields. Many of the most troublesome weeds for contemporary maize farmers in the northeast are European imports, which arrived after the introduction of European crops and animals: hedge and field bindweed (*Convolvulus arvensis* L. and *C. sepium* L.); redroot pigweed (*Amaranthus retroflexus* L.); chickweed (*Stellaria media*); purslane (*Portulaca oleracea* L.); quackgrass (*Elytrigia repens*); shepherd's purse (*Capsella bursa-pastoris* L.); sow thistle (*Sonchus arvensis* L.); black nightshade (*Solanum nigrum* L.); dandelion (*Taraxacum officinale*); narrow and broadleaved plantain (*Plantago lanceolata* L. and *P. major* L.); and crabgrass (*Digitaria sanguinalis*). These weeds moved rapidly into the Atlantic coastal colonies and New England following the arrival of Europeans in the seventeenth century. Most non-native weeds gained their footholds in New York fields through dispersal by animal manure and subsequent incorporation with plows. In central and western New York, Seneca farmers maintained control of their territories until the close of the Revolutionary War, preventing colonists and their weeds from moving into their lands. Although some domesticated animals were found in Seneca territories during the colonial era, there are no records that they were used as draft animals or that their manure was applied to agricultural fields. As a result, European weeds did not invade Seneca country until after Sullivan's campaign in 1787, leaving their maize fields relatively free of these weedy pests until after the Revolutionary War.³⁷

Comparing labor requirements in two very different agricultural systems across two continents is fraught with problems. First, the economic, political, and cultural systems under which farmers in the two areas worked differed markedly. While European farmers produced grain for subsistence, under incipient capitalism they were also part of a market system that determined the value of their labor and other resources. Iroquois farmers, in contrast, operated largely outside of a market system, with the value of labor and its products determined by a very different metric. Differences in the food distribution systems as well as gender roles in farm labor further complicate comparisons. Second, while there is some data on European labor requirements for farm production

during this time, comparative data for Iroquois agriculture is completely absent.

However, it is possible to speculate on primary differences in the amount of labor for the two agricultural systems. Plow-based wheat production would seem to require more labor than maize. First, caring for animals, including feeding and manure management, added many additional chores to daily work that Iroquois maize farmers did not have. Second, wheat farmers plowed and harrowed their fields to prepare them for planting, requiring multiple trips across the field. Iroquois farmers prepared the field and planted in a single operation, removing vegetation from the hills as they planted the corn kernels. But they had to return to their fields to weed at least once or twice, and children were often posted in fields early and late in the growing season to keep birds away. Because wheat cannot be weeded, European farmers invested labor in weed control efforts in the row crop that preceded wheat, but they did not weed their fields. An anecdotal report suggests that the work of growing Iroquois maize was not terribly burdensome. Mary Jemison, a white woman who lived most of her life with the Seneca, said of her agricultural work, "In the summer season, we planted, tended and harvested our corn, and generally had all our children with us; but had no master to oversee or drive us, so that we could work as leisurely as we pleased." It seems unlikely that European farmers in this time period would have described their labor as "leisurely."³⁸

Thus, the Iroquois produced considerably more grain per acre than European farmers. But some of this yield differential is attributable to the biological characteristics of maize itself. After the Revolutionary War, American farmers removed the Seneca and established their own plow-based systems across western New York on lands that had been previously unplowed. The record of maize yields by American farmers in this region in the nineteenth century provides insight into the effects of tillage alone.

After the defeat of the Iroquois Confederacy in the American Revolution, land speculators moved in to buy up large swaths of land in western New York that had been held by the Seneca and Cayuga nations. They sold it to US farmers who had been pressuring English and American colonial governments for access to these lands for almost half a century. Soldiers who participated in Sullivan's Campaign had seen first-hand the

agricultural potential of the region, and reports by speculators and visitors provided detailed descriptions of fertile soils, vast tracts of timber, and other resources. As a result, many farmers were eager to settle the Genesee Country, which included land west of Seneca Lake, between Pennsylvania and Lake Ontario. In 1810 this area had a population of slightly more than 23,000, but by 1860 the population had increased to almost 265,000. These farmers brought plows and draft animals to western New York, tilling these lands for the first time. Although most farmers intended to grow wheat, the primary cash crop of the time, maize was an agricultural staple. After clearing land, farmers first planted maize to feed themselves and their animals and only then cleared additional acreage to plant wheat and other crops.³⁹

Beginning in 1792 speculators and other visitors to western New York provided assessments for the incoming settlers regarding the region's potential for agriculture. In a series of letters written between 1792 and 1798, an anonymous visitor to the Genesee Country, reporting on lands in its northern half, stated, "the Soil is remarkably rich" and that it was "the best in the world for raising large and productive crops of hemp, flax, indian corn." He indicated that farmers could expect excellent crops, particularly in the fertile river bottoms, with estimated maize yields of thirty to sixty bushels per acre. In the flats along the Genesee River, he wrote, "the Indians raise one hundred bushels of corn on an acre." In 1804 Robert Munro, a land speculator in western New York, wrote, "The quality of the soil is various, but in the better or most even parts of the country before mentioned, a rich loamy soil is the most common, and it is sometimes covered on the top with a loose black mould, from six to ten inches deep." In terms of expected crop yields, Munro wrote, "Corn generally grows to the amount of 30 to 40 bushels on an acre, and in some instances, on the flats of the Genesee river and of Mud Creek, it is said, to the extent of 70 to 80 bushels an acre, and is very heavy and substantial in quality."⁴⁰

Beginning in the nineteenth century, as part of the federal census, farmers recorded total production for cereal grains in each of New York's counties. But it was not until 1845 that they first reported the number of acres planted to maize and the amount of harvested grain in each county, enabling the calculation of yields per acre. Table 6 provides mean maize yields for five counties in the Genesee Country: Genesee, Livingston,

Monroe, Ontario, and Seneca. In 1845 and 1855 maize yields averaged just twenty-eight and twenty-six bushels per acre, respectively. Only in 1880 did yields first climb above forty bushels per acre; they reached fifty bushels per acre in 1920 and ranged between thirty and forty bushels per acre for all other censuses in this period.

Yields of twenty-six and twenty-eight bushels per acre in 1845 and 1855 were mediocre at best, considerably lower than what first settlers had expected or obtained in the early nineteenth century, and well below those of Seneca farmers in the seventeenth and eighteenth centuries. What accounts for the failure of US farmers to produce bumper harvests of maize in a region noted for its extraordinary soil fertility? And why did yields eventually increase to forty to fifty bushels per acre?

By 1855 much of the region had been under plow agriculture for forty years or more. Farmers plowed their maize fields and then harrowed them at least once to form a smooth seedbed before planting. Once the

Table 6. Average Maize Yields in Five Western New York Counties, 1845–1940.

	<i>Genesee</i>	<i>Livingston</i>	<i>Monroe</i>	<i>Ontario</i>	<i>Seneca</i>	<i>Mean</i>
<i>Year</i>	<i>Yield bu acre¹</i>					
1845	27	26	30	28	27	28
1855	27	28	27	26	24	26
1865	33	34	34	39	37	35
1875	39	39	35	37	38	38
1880	42	41	44	42	41	42
1890	37	34	32	35	42	36
1900	28	31	29	32	31	30
1910	42	39	39	32	31	37
1920	50	50	47	50	51	50
1930	43	38	41	40	40	40
1940	40	39	42	40	35	39

SOURCE: New York State Secretary's Office, *Census of the State of New York, for 1845* (Albany: Carroll & Cook, 1846); New York State Secretary's Office, *Census of the State of New York, for 1855* (Albany: C. van Benthuyssen, 1857); New York State Secretary's Office, *Census of the State of New York, for 1865* (Albany: C. van Benthuyssen, 1867); New York State Secretary's Office, *Census of the State of New York, for 1875* (Albany: Weed, Parsons and Co. Printers, 1877); US Census Office, *United States Census Reports on Agriculture* (Washington, DC: GPO, 1880, 1890, 1900, 1910, 1920, 1930, 1940).

crop was up, they also cultivated, usually two or three times, to remove weeds between the rows. Most farmers grew maize, wheat, and other row crops (potatoes, turnips, beans) continuously, year after year. The combination of these practices that included intensive tillage operations and continuous production of row crops, resulted in steady oxidation of soil organic matter, decreased fertility, and lower crop yields, the same problems that European farmers faced with their plow-based wheat production in the seventeenth and eighteenth centuries. Farmland that had once been very fertile, within several decades of continuous tillage, was producing mediocre maize yields.⁴¹

But not all farmers in the nineteenth century engaged in these destructive practices. There are many accounts of farmers in New York and elsewhere in the country growing maize that yielded sixty to as much as two hundred bushels per acre. In every case, these growers had learned to increase soil organic matter and fertility in fields that had been intensively tilled by establishing crop rotations with forage legumes (clover, alfalfa) and by applying large amounts of animal manure. Although neither agricultural scientists nor farmers at this time identified the role that tillage played in the loss of soil organic matter, all agreed that good stewardship and high-yielding crops in plow-based agriculture required the use of forage legumes, rotations, and manure. Over many decades and generations, growers began to adopt these practices; average maize yields in these five counties gradually increased to levels that Seneca farmers had obtained centuries earlier with their no-plow agriculture.⁴²

Iroquois farmers in the seventeenth and eighteenth centuries produced more grain per unit land area than their European counterparts for two reasons: by not plowing their fields, the Iroquois conserved soil organic matter that ensured a stable supply of plant-available nitrogen, while European plow-based agriculture oxidized soil organic matter, which limited wheat yields because of insufficient nitrogen; and maize characteristics, including its larger seed size, greater seedling and plant vigor, lower seeding rate, higher yield seed ratios, and higher yield potential, provided inherent yield advantages over wheat.

Historians and other scholars have overstated and misunderstood the role of tillage in agriculture, attributing to it advances in agricultural development, but failing to recognize the inherent weaknesses associated

with permanent cropping systems that depend on plowing. Continuous tillage in Europe in the seventeenth and eighteenth centuries reduced soil organic matter to levels where insufficient nitrogen limited grain production to less than fifteen bushels per acre. Plow-based systems in Europe did not overcome this problem until farmers introduced forage legumes into their cropping systems in the nineteenth century and later. Iroquois farmers on the fertile soils of the Ontario Lake Plain in the seventeenth and eighteenth centuries developed a stable, continuous cropping system that maintained soil organic matter levels by *not plowing*. Their soil-conserving practices, today referred to as no-till or conservation tillage, provided sufficient nitrogen to produce forty to seventy-five bushels per acre of grain. Iroquois farmers also had the advantage of a more productive cereal grain, maize, with its higher yield potential and higher yield seed ratio than wheat. In Iroquois cropping systems maize had yield seed ratios of between 154 to 320, compared to 4 to 21 for European wheat. Seeding rates for maize ranged from eight to fifteen bushels per acre, but wheat required sixty to one hundred eighty bushels per acre. Consequently, wheat farmers had lower net yields because they had to save a higher portion of their harvest for planting. Maize is also less vulnerable to losses from pests and is uniquely well-being to roughly prepared seedbeds that are characteristic of no-till agriculture.

Although maize yields are usually higher than wheat in any given environment and cropping system, tillage explains a significant portion of the yield advantage of the Iroquois over Europeans. United States settlers moved into Seneca territories in the nineteenth century and began to grow maize using plow culture. In less than fifty years maize yields in the fertile Genesee Country of western New York decreased to less than thirty bushels per acre. These farmers experienced the same problems as European wheat farmers who had continuously plowed their fields, with the subsequent oxidation of soil organic matter and plummeting crop yields. Maize yields in western New York rebounded in the latter part of the nineteenth century to levels achieved by the Iroquois more than two hundred years earlier when farmers implemented crop rotations with forage legumes and applied animal manure to their plowed fields to restore soil organic matter lost through plowing.

NOTES

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